

3. (a) Give the definition of the *specific heat capacity* of a material. [1]

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- (b) An insulating flask contains $0.6 \times 10^{-3} \text{ m}^3$ of water at 19.5°C . A volume $1.6 \times 10^{-3} \text{ m}^3$ of boiling water at 100.0°C is poured into the flask.

(specific heat capacity of water, $c = 4\,200 \text{ J kg}^{-1}^\circ\text{C}^{-1}$; density of water, $\rho = 1\,000 \text{ kg m}^{-3}$)

- (i) Determine the final temperature of the water in the flask. [3]

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- (ii) Calculate the heat lost by the boiling water. [3]

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- (iii) Justify the statement that no work is done on/by the boiling water as it cools. No further calculations are needed. [1]

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